

Lecture 6: Ito Calculus & Ito's Lemma

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Based on lecture notes by me and many previous CS335 instructors

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Recall: Properties of Brownian Motion

As $\Delta t \rightarrow 0$, the distance travelled by a particle in a *finite* time becomes *infinite*. Why? Well, for one lattice step we know:

$$|X_{i+1} - X_i| = |\Delta X| = \Delta h$$

Then total (absolute) distance travelled in time t (over n steps) is

$$n|\Delta X| = \frac{t}{\Delta t} \sigma \sqrt{\Delta t} = \frac{t\sigma}{\sqrt{\Delta t}} \rightarrow \infty \quad \text{as } \Delta t \rightarrow 0.$$

Similarly, as $\Delta t \rightarrow 0$,

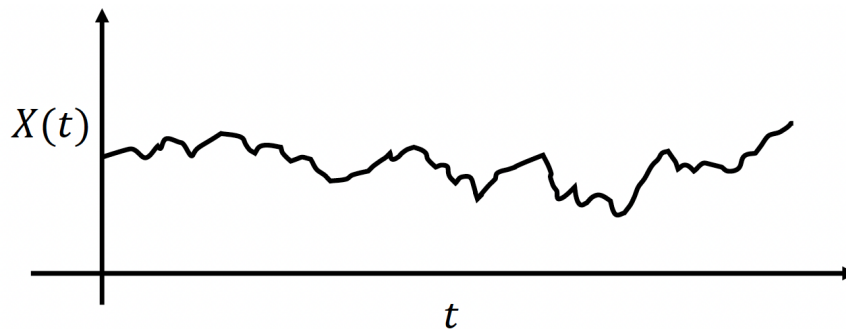
$$\frac{\Delta X}{\Delta t} = \pm \frac{\sigma \sqrt{\Delta t}}{\Delta t} = \pm \frac{\sigma}{\sqrt{\Delta t}} = \pm \infty$$

So, this attempt to approximate the slope over a finite time period returns an answer of *infinity*. Uh on...

Brownian motion and Ito calculus

What is this telling us? Well, these paths are very “jagged” at *every* time scale, regardless of how far you “zoom in”.

The paths taken by $X(t)$ are non-differentiable!



“Regular” calculus derivatives can’t be safely applied - to deal with them, we will need ideas from Ito/Stochastic Calculus.

Ito Calculus

Consider the SDE $dX = \alpha(X, t)dt + c(X, t)dZ$, recalling that $dZ = \phi\sqrt{dt}$ (e.g., X might be a stock price).

We interpret the SDE as expressing an integral over a short time from 0 to t :

$$X(t) - X(0) = \int_0^t dX(s) = \int_0^t \alpha(X(s), s)ds + \int_0^t c(X(s), s)dZ(s).$$

Notice we are integrating over time, but the last integral expression involves dZ (a function of dt) rather than dt directly.

What do we do with this?

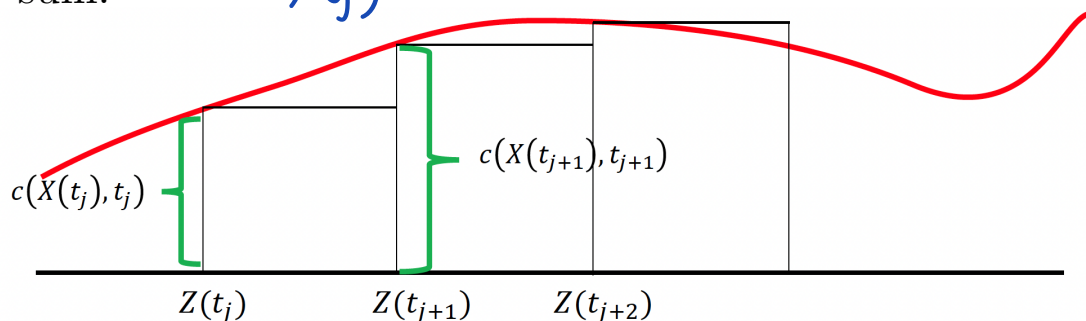
Ito Stochastic Integral

Notation: Let $\Delta Z(t_j) = Z(t_{j+1}) - Z(t_j)$ and time step $\Delta t = t_{j+1} - t_j$.

We define the (Ito) stochastic integral as

$$\int_0^t c(X(s), s) dZ(s) = \lim_{\Delta t \rightarrow 0} \sum_{j=0}^{N-1} c_j \Delta Z_j$$

where $c_j = c(X(t_j), t_j)$ and $N = \frac{t}{\Delta t}$. i.e., the integral is the limit of a discrete sum.



Note: we are using the height (c value) at the *left end* of each interval!

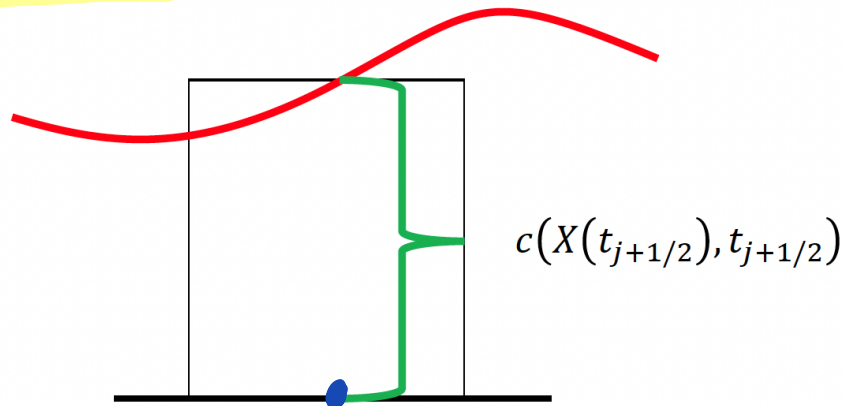
This is the form used by Ito Calculus.

Alternative: Stratonovich rule

Alternatively, one *might* image using e.g., “mid-step” heights, like:

$$\int_0^t c(X(s), s) dZ(s) = \lim_{\Delta t \rightarrow 0} \sum_{j=0}^{n-1} c(X(t_{j+1/2}), t_{j+1/2}) dZ_j$$

with $t_{j+1/2} = (t_{j+1} + t_j)/2$. Called the “Stratonovich rule”.



No peeking!

In finance, we use the Ito stochastic calculus, rather than this Stratonovice rule. Why?

We *don't yet know* the values (e.g., of the stock) at the future point in time, $t_{j+1/2}$, so we should not use them in making decisions.

Otherwise, we would be “peeking into the future” when making decisions, which is **forbidden!** Our mathematics should reflect this.

Ito's lemma

A key result we'll need from Ito calculus is called
Ito's lemma.

It is, in essence, a generalization of the familiar chain rule for stochastic differential equations.

We will now work towards this result.

A Useful Rule

A few slides from now, we will need to interpret $\int_0^t c(X(s), s) (dZ(s))^2$.

To do so, we will make use of the following claim:

$$\int_0^t c(X(s), s) (dZ(s))^2 = \int_0^t c(X(s), s) ds$$

or equivalently

$$\lim_{\Delta t \rightarrow 0} \sum_{j=0}^{n-1} c_j (dZ_j)^2 = \lim_{\Delta t \rightarrow 0} \sum_{j=0}^{n-1} c_j \Delta t$$

We will interpret this as implying a useful rule of Ito calculus, that:

$$dZ^2 = dt \quad \text{as} \quad dt \rightarrow 0$$

A Useful Rule

$$dZ^2 = dt \quad \text{as} \quad dt \rightarrow 0$$

This should seem at least intuitively reasonable. Why?

We defined $dZ = \phi\sqrt{dt}$, thus $dZ^2 = \phi^2 dt$ and $E[\phi^2] = 1$, so $E[dZ^2] = dt$. i.e., it holds in expectation.

A deeper exploration of this can be found in the course notes (in the Chapter 2 material).

Big-O notation – infinite case

Another tool we will need is Big-O notation, which describes asymptotic behaviour of polynomials, $p(x)$, in two settings: $x \rightarrow 0$ and $x \rightarrow \infty$.

We say that $p(x) = O(x^n)$ as $x \rightarrow \infty$ if there exists $C > 0, x_0 > 0$ such that $|p(x)| < C |x^n|$ for all $x > x_0$.

e.g. $p(x) = 7x^{10} + 4x^{11} = O(x^{11})$ as $x \rightarrow \infty$

Thus, in the limit, the x^{11} term dominates over the other term. (Course notes provide additional examples/details.)

Big-O notation – infinitesimal case

We say that $p(x) = O(x^n)$ as $x \rightarrow 0$ if there exists $C > 0, \delta > 0$ such that $|p(x)| < C|x^n|$ for all $|x| < \delta$.

e.g. $p(x) = 7x^{10} + 4x^{11} = O(x^{10})$ as $x \rightarrow 0$

$$x = 0.001$$

This notation lets us characterize how polynomials behave in the limit, i.e., which terms remain important when x gets large or small (without necessarily worrying about the actual coefficients).

Big-O notation - arithmetic

We can also do some basic arithmetic with this notation!

e.g. If $f(x) = x + O(x^2)$ and $g(x) = 2x + O(x^3)$, then as $x \rightarrow 0$:

$$f(x) + g(x) = 3x + O(x^2) + \cancel{O(x^3)} = 3x + O(x^2) = O(x)$$

and

$$\begin{aligned} f(x) \times g(x) &= 2x^2 + \cancel{2x \times O(x^2)} + \cancel{x \times O(x^3)} + O(x^5) \\ &= 2x^2 + \downarrow O(x^3) + \downarrow O(x^4) + O(x^5) \\ &= 2x^2 + \cancel{O(x^3)} = \cancel{O(x^2)} \end{aligned}$$

Back to Ito Calculus

Suppose we have a variable X undergoing Brownian motion (as usual),

$$dX = a(X, t)dt + b(X, t)dZ \quad \text{with} \quad dZ = \phi\sqrt{dt}, \quad \phi \sim N(0, 1)$$

and F is a function depending on X and t , i.e., $F(X(t), t)$.

How should F evolve?

As $t \rightarrow t + dt$ and $F \rightarrow F + dF$, then a Taylor series implies that...

$$dF = \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial X} dX + \frac{1}{2} \frac{\partial^2 F}{\partial X^2} (dX)^2 + \dots$$

Let's work out dX^2 and apply Big- O to simplify!

Deriving Ito's Lemma

We can multiply through to find that:

$$(dX)^2 = (adt + bdZ)^2 = a^2 dt^2 + 2ab dt \cdot dZ + b^2 (dZ)^2$$

and we remember that:

$$dZ = \phi \sqrt{dt} = O(dt^{0.5}) \quad \text{as } dt \rightarrow 0$$

$$(dZ)^2 = dt \quad \text{as } dt \rightarrow 0 \quad \leftarrow \text{from slide 9}$$

Therefore $(dX)^2 = a^2 dt^2 + 2ab \times O(dt^{1.5}) + b^2 dt$

Or, again applying Big- O (assuming small dt),

$$(dX)^2 = b^2 dt + O(dt^{1.5})$$

Now let's go back to our expression for dF ...

Deriving Ito's Lemma

Now:

$$\begin{aligned} dF &= F_t dt + F_x dX + \frac{1}{2} F_{xx} (dX)^2 + \dots \\ &= F_t dt + F_x (adt + bdZ) + \frac{1}{2} F_{xx} (b^2 dt) + O(dt)^{1.5} \end{aligned}$$

Grouping terms:

$$= bF_x dZ + \left(aF_x + \frac{b^2}{2} F_{xx} + F_t \right) dt + O(dt)^{1.5}$$

We again have a drift term and a random term, but also one more that is “small” (i.e., goes to zero).

End of Lecture 6

A note on the last (“small”) term

Remember we interpreted our SDEs as describing integrals, so for dF we have:

$$\int_0^t dF = \int_0^t \left(bF_X dZ + \left(aF_X + \frac{b^2}{2} F_{XX} + F_t \right) dt \right) + O \left(\int_0^t dt^{1.5} \right)$$

Consider the last term as the limit of a discrete sum:

$$\int_0^t dt^{1.5} = \lim_{\Delta t \rightarrow 0} \sum_0^{N-1} (\Delta t)^{1.5} =$$

Thus, it does not contribute in the limit.

Ito's Lemma

We find the following result, called _____:

If $dX = a(X, t)dt + b(X, t)dZ$ and $F = F(X, t)$, then

$$dF = \left(aF_X + \frac{b^2}{2}F_{XX} + F_t \right) dt + bF_X dZ$$

or

$$F(t) - F(0) = \int_0^t \left(aF_X + \frac{b^2}{2}F_{XX} + F_t \right) dt + \int_0^t bF_X dZ$$

This resembles the chain rule but for *stochastic* (random) processes.

So e.g., if we have a stock price X evolving via an SDE, and another variable F that depends on that stock's value, we can use this to determine the SDE describing how F evolves.